Confidence-Gated Dynamic Hierarchical Detection and Information-Theoretic Concept Drift Tracking for Out-of-Distribution Network Intrusion Detection

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***Abstract*—Modern Network Intrusion Detection Systems (NIDS) deployed in high-speed enterprise perimeters face a fundamental architectural trade-off: traditional fast-path rule engines process packet headers at multi-gigabit speeds but lack discriminatory depth against stealthy exploits, whereas deep Machine Learning (ML) ensembles achieve high in-distribution classification accuracy but suffer performance degradation under out-of-distribution (OOD) attack streams while imposing heavy computational latency. Furthermore, standard supervised ML models cannot quantify their own predictive uncertainty when distribution shifts occur. To resolve these challenges, this paper presents a novel Confidence-Gated Dynamic Hierarchical Detection (CG-DHD) framework coupled with an Information-Theoretic Concept Drift Tracking Engine. Our architecture decouples network telemetry into a Tier-1 Fast-Path EWMA Heuristic Filter (0.08 μs per packet-equivalent Tier-1 header operation, ≈ 12.5 million packet evaluations/sec (12.5 Mpps); this Mpps figure represents the reciprocal of the Tier-1 operation latency rather than full pipeline throughput) and a Tier-2 Stateful Sliding-Window Ensemble (Random Forest + XGBoost, full CG-DHD pipeline latency 20.63 μs/flow). We introduce a formal mathematical arbitration model where high-confidence ML predictions suppress heuristic false alarms, whereas predictive entropy surges $H(y|x)$ condition a dual ADWIN and Page-Hinkley streaming drift monitor. Evaluated across 108,000 network flows spanning CICIDS-2017, CIC-DDoS2019, and UNSW-NB15, including zero-shot cross-dataset transfer testing on CIC-DDoS2019 (serving as an empirical proxy for OOD attack conditions), our system detects novel OOD attack onset within 111–399 flows (< 70 ms) without prior exposure. Rather than targeting marginal gains in static classification accuracy over standalone models, CG-DHD provides a favorable system-level Pareto trade-off, achieving an in-distribution accuracy of 99.88%, a low false alarm rate of 0.14%, and sub-70 ms drift detection delay. Hardware throughput profiling demonstrates an effective processing rate of 48,483 flows/sec (4.77 Gbps equivalent) with a memory footprint under 781 MB. An analytical Hoeffding derivation yields the sub-window cut threshold for confidence parameter $\delta = 0.002$, with empirical validation confirming zero false drift alarms across 50,000 monitored telemetry windows.**

***Index Terms*—Network Intrusion Detection, Out-of-Distribution (OOD) Traffic, Concept Drift, Shannon Predictive Entropy, Hierarchical Machine Learning, ADWIN, Hoeffding Bound, Hardware Throughput Profiling, eBPF/XDP Offloading.**

I. Introduction

With the continuous escalation of enterprise multi-cloud infrastructures, modern software-defined networks, and high-throughput data centers, the volume, velocity, and architectural complexity of network traffic have expanded exponentially [21], [24]. Enterprise security perimeters are tasked with ingesting, inspecting, and classifying millions of network flows per second under strict high-throughput and low-latency constraints. Concurrently, the global cyber threat landscape has witnessed a rapid proliferation of sophisticated Distributed Denial of Service (DDoS) amplification vectors, multi-stage advanced persistent threats (APTs), and novel exploit payloads [11]. In this adversarial environment, Network Intrusion Detection Systems (NIDS) serve as the fundamental defensive backbone for identifying malicious telemetry, safeguarding critical infrastructure, and ensuring operational availability.

However, existing commercial and open-source NIDS architectures are constrained by a deep architectural dichotomy:

1) **Deterministic Rule Engines (Deep Packet Inspection):** Fast-path signature filters (e.g., Snort [12], Suricata [13], Bro/Zeek [14]) utilize hand-crafted rule databases and packet header thresholds. Operating directly on raw packet streams, these engines execute with microsecond latency (< 0.1 μs per packet) and scale to tens of gigabits per second. Nevertheless, deterministic rule engines suffer from two crippling flaws: complete inability to detect novel out-of-distribution (OOD) exploits or obfuscated payload mutations, and high false positive rates (FAR $> 30\%$) under bursty benign network activity.
2) **Statistical & Deep Machine Learning Classifiers:** Supervised machine learning algorithms (Random Forest [22], XGBoost [4], Deep Neural Networks [19], Autoencoders [8]) extract rich multi-dimensional flow features to identify non-linear anomaly patterns. While achieving near-perfect in-distribution accuracy ($> 99.9\%$) on static benchmark datasets (such as CICIDS-2017 [1]), supervised ML models can experience substantial performance degradation when exposed to out-of-distribution (OOD) traffic streams [20].

The vulnerability of supervised ML classifiers under OOD network conditions stems from their greedy decision boundary partitioning. Standard decision trees and neural networks assign deterministic class label predictions without measuring model calibration or boundary ambiguity. Consequently, when an unseen out-of-distribution reflection attack (e.g., DrDoS_DNS or TFTP amplification [11]) arrives, supervised classifiers fail silently, outputting overconfident benign classifications and causing severe recall degradation under distribution shifts. Furthermore, evaluating heavy tree ensembles or multi-layer neural networks imposes a severe processing latency bottleneck (3–18 μs per flow), causing buffer conges-

tion, packet dropping, and throughput degradation under multi-gigabit traffic rates [9]. In this context, cross-dataset transfer testing provides a rigorous empirical proxy for evaluating OOD attack adaptation under realistic network distribution shifts.

A. Threat Model and Operational Boundaries

The proposed CG-DHD architecture is engineered to defend high-speed enterprise network perimeters against non-stationary volumetric and zero-day exploit streams under realistic operational constraints:

- **In-Scope Threat Vectors:** Volumetric DDoS amplification attacks (DrDoS_DNS, DrDoS_NTP, TFTP, SSDP, SNMP), TCP SYN floods, high-rate probing, and unseen out-of-distribution (OOD) exploit payloads originating from external perimeters.
- **Out-of-Scope Threat Vectors:** Low-and-slow evasive exfiltration (e.g., sub-threshold covert timing channels), training data poisoning attacks, and hardware side-channel attacks on host memory.

To overcome this fundamental dilemma, this paper proposes **Confidence-Gated Dynamic Hierarchical Detection (CG-DHD)**, an advanced multi-tier architectural framework that decouples network telemetry processing into a microsecond Tier-1 Fast-Path EWMA Heuristic Filter and a Tier-2 Stateful Sliding-Window ML Ensemble. Rather than competing solely on static in-distribution classification accuracy—where standalone Random Forest or XGBoost already reach $> 99.9\%$—CG-DHD targets system-level Pareto efficiency across accuracy, false-positive attenuation, inference throughput, and out-of-distribution concept drift detection. To bridge the gap between static model classification and dynamic traffic evolution, we integrate an **Information-Theoretic Concept Drift Tracking Engine** that monitors Normalized Shannon Predictive Entropy $H(y|x)$ [6] over streaming model predictions.

Rather than relying on static decision thresholds, CG-DHD treats predictive entropy as a continuous real-time signal of classification ambiguity. By feeding streaming entropy values into a dual Adaptive Windowing (ADWIN [3]) and Page-Hinkley [16] drift detector (in contrast to supervised error-rate monitors such as DDM [17] and EDDM [18] that require immediate true labels), our system employs an analytically derived sub-window cut threshold based on Hoeffding's Inequality [7] ($\delta = 0.002$), validated empirically against bursty telemetry streams. When unseen OOD traffic induces a sudden surge in predictive entropy, the system dynamically flags suspected OOD attack streams and triggers retraining alerts for downstream model updating within sub-70 millisecond bounds (111–399 flows). Simultaneously, when Tier-2 ML confidence is high ($1 - H(y|x) \geq 0.70$), a Confidence-Gated Arbitration function suppresses false heuristic alerts generated by Tier-1 under noisy benign traffic, suppressing system-wide false positive rates to 0.14%.

B. Research Objectives & Questions

This investigation resolves four central research questions:

- **RQ1 (Throughput vs. Accuracy):** Can a decoupled two-tier architecture achieve 0.08 μs packet-equivalent Tier-1 operation latency while maintaining stateful ensemble classification accuracy ($> 99.8\%$) in the deeper detection plane?
- **RQ2 (Uncertainty Quantification):** How effectively does Normalized Shannon Predictive Entropy $H(y|x)$ quantify classification ambiguity under unseen out-of-distribution (OOD) attack vectors?
- **RQ3 (Mathematical Drift Bounds):** Can streaming concept drift monitoring over predictive entropy provide a mathematically bounded false-alarm probability under stationary no-drift conditions, while maintaining low empirical false-alarm rates under bursty network streams?
- **RQ4 (Arbitration & Hardware Efficiency):** How does dynamic arbitration between Tier-1 rate heuristics and Tier-2 prediction confidence affect operational false positive rates and hardware memory/CPU resource consumption?

C. Summary of Scientific Contributions

The main scientific contributions of this research are summarized as follows:

- **Decoupled Two-Tier Hybrid Architecture (CG-DHD):** We present a multi-tier NIDS design combining a 0.08 μs packet-equivalent Tier-1 EWMA rate operation with a stateful 57-feature ML ensemble (Random Forest [22] + XGBoost [4]), addressing the latency–accuracy trade-off.
- **Information-Theoretic Uncertainty Metric:** We establish Normalized Shannon Predictive Entropy $H(y|x)$ [6] over ensemble posterior distributions as a mathematically defined indicator of model ambiguity during zero-shot OOD traffic shifts.
- **Analytical Derivation & Empirical False-Alarm Bound:** We provide a formal analytical derivation of the ADWIN sub-window cut threshold using Hoeffding's Inequality [7] ($\delta = 0.002$) and empirically verify false-alarm suppression ($0/50,000$ false alarms) under auto-correlated streaming traffic.
- **Confidence-Gated Arbitration Function:** We formulate a dynamic arbitration model that suppresses noisy fast-path heuristic alerts when ML prediction certainty is high ($1 - H(y|x) \geq 0.70$), reducing system false alarm rates to 0.14%.
- **Rigorous Benchmark & Hardware Validation:** We validate CG-DHD across 108,000 network flows spanning CICIDS-2017 [1], CIC-DDoS2019 [11] (11 distinct unseen OOD attack streams), and UNSW-NB15 [10] datasets. Hardware throughput profiling confirms an effective throughput of 48,483 flows/sec (4.77 Gbps bandwidth equivalent) with a memory RSS footprint below 781 MB.

II. Formal Mathematical Formulation & Analytical Proofs

In this section, we formulate the mathematical algebra governing bidirectional flow aggregation, information-theoretic

uncertainty quantification, streaming concept drift detection bounds, and decision arbitration logic.

A. Feature Space & Bidirectional Flow Algebra

Let $\mathcal{S} = \{p_1, p_2, \ldots, p_N\}$ denote a continuous stream of network packets arriving at a network interface. Each packet $p_i = (t_i, s_ip_i, d_ip_i, s_prt_i, d_prt_i, proto_i, len_i, flags_i)$ is mapped to a unique 5-tuple key:

$$K(p_i) = \Big(\min(s_ip, d_ip), \max(s_ip, d_ip), \min(s_prt, d_prt), \max(s_prt, d_prt), proto \Big) \tag{1}$$

Packets sharing 5-tuple key K within a sliding time window $\tau = 5.0$ seconds constitute a stateful flow sequence $F_K = (p_{K,1}, p_{K,2}, \ldots, p_{K,m})$. The feature extraction function $\Phi : F_K \to \mathbb{R}^d$ computes a d-dimensional feature vector $x \in \mathcal{X} \subset \mathbb{R}^{57}$, containing 47 standard statistical metrics (duration, inter-arrival time statistics, packet length variance) and 10 custom engineered ratio features (formally defined in Table III).

B. Information-Theoretic Predictive Entropy

Let $\mathcal{Y} = \{1, 2, \ldots, C\}$ represent the set of C discrete attack target classes. The Tier-2 stateful ensemble model computes a posterior probability distribution $P(y = c \mid x)$ over all classes $c \in \mathcal{Y}$, satisfying:

$$\sum_{c=1}^{C} P(y = c \mid x) = 1, \quad P(y = c \mid x) \geq 0, \quad \forall c \in \mathcal{Y} \tag{2}$$

To measure model uncertainty when evaluating flow vector x, we formulate Normalized Shannon Predictive Entropy $H(y \mid x)$.

Definition 1 (Normalized Shannon Predictive Entropy)**.** *For an observed flow feature vector $x \in \mathcal{X}$, the Normalized Shannon Predictive Entropy $H(y \mid x) \in [0, 1]$ is defined as:*

$$H(y \mid x) = -\frac{1}{\log_2 C} \sum_{c=1}^{C} \hat{P}(y = c \mid x) \log_2 \hat{P}(y = c \mid x) \tag{3}$$

where $\hat{P}(y = c \mid x) = \frac{\max(P(y=c|x), \epsilon)}{\sum_{j=1}^{C} \max(P(y=j|x), \epsilon)}$ with clipping parameter $\epsilon = 10^{-12}$ to enforce numerical stability while guaranteeing probability normalization ($\sum \hat{P} = 1$), and normalization constant $\log_2 C$ rescales the metric strictly onto the unit interval $[0, 1]$.

Lemma 1 (Mathematical Bounds of Normalized Entropy)**.** *For any C-class probability distribution $P(y \mid x)$, the Normalized Shannon Entropy satisfies $0 \leq H(y \mid x) \leq 1$.*

1) *Minimum Entropy ($H(y \mid x) = 0$): Holds if and only if $\exists k \in \mathcal{Y}$ such that $P(y = k \mid x) = 1$ and $P(y = c \mid x) = 0$ for all $c \neq k$. This represents absolute classification certainty.*
2) *Maximum Entropy ($H(y \mid x) = 1$): Holds if and only if $P(y = c \mid x) = \frac{1}{C}$ for all $c \in \mathcal{Y}$. This represents complete predictive ambiguity and may occur under severe distributional mismatch, including OOD conditions.*

Proof. By Jensen's Inequality, since $f(t) = -t \log_2 t$ is strictly concave for $t > 0$, the sum $\sum_{c=1}^{C} -P(c) \log_2 P(c)$ attains its unique maximum when $P(c) = \frac{1}{C}$ for all c. Evaluating at equality yields $\sum_{c=1}^{C} -\frac{1}{C} \log_2 \left(\frac{1}{C}\right) = \log_2 C$. Dividing by $\log_2 C$ yields the upper bound 1. Conversely, as $P(k) \to 1$, $1 \cdot \log_2(1) = 0$, yielding the lower bound 0. □

C. Adaptive Windowing (ADWIN) & Analytical Drift Thresholding

To detect non-stationary concept drift across streaming predictive entropy observations $H_1, H_2, \ldots, H_t \in [0, 1]$, we employ an Adaptive Windowing (**ADWIN**) algorithm. ADWIN dynamically adjusts the length of a sliding memory window W based on statistical variation.

Definition 2 (ADWIN Sub-window Cut Criterion)**.** *Let W be a window of recent entropy observations of length $|W|$. For any binary partition of W into two contiguous sub-windows W_0 and W_1 ($W = W_0 \cdot W_1$), let $\hat{\mu}_0$ and $\hat{\mu}_1$ denote the sample means of entropy in W_0 and W_1, with sizes $n_0 = |W_0|$ and $n_1 = |W_1|$ respectively. Concept drift is declared if:*

$$|\hat{\mu}_0 - \hat{\mu}_1| > \epsilon_{\text{cut}} \tag{4}$$

where the adaptive cut threshold ϵ_{cut} is derived via Hoeffding's bound as:

$$\epsilon_{\text{cut}} = \sqrt{\frac{1}{2m} \ln \left(\frac{4|W|}{\delta} \right)} \tag{5}$$

where $m = \left(\frac{1}{n_0} + \frac{1}{n_1} \right)^{-1} = \frac{n_0 n_1}{n_0 + n_1}$ is the harmonic mean of sub-window sample sizes, and $\delta \in (0, 1)$ is the confidence parameter.

Proposition 1 (Hoeffding Bound Motivation for Fixed Sub--Partitions)**.** *For a fixed window W containing independent entropy random variables $H_i \in [0, 1]$ generated under stationary null distribution H_0 ($\mu_0 = \mu_1$), Hoeffding's Inequality bounds the probability of a false sub-window cut across any single fixed partition by:*

$$P_{H_0} \left(|\hat{\mu}_0 - \hat{\mu}_1| \geq \epsilon_{\text{cut}} \right) \leq 2 \exp \left(-2\, m \cdot (\epsilon_{\text{cut}})^2 \right) = \frac{\delta}{2|W|} \tag{6}$$

Applying a union bound across all $|W| - 1$ candidate binary sub-partitions of a fixed window yields $P_{H_0}(\exists W_0, W_1 : |\hat{\mu}_0 - \hat{\mu}_1| > \epsilon_{\text{cut}}) < \delta/2 < \delta$.

Remark 1 (Sequential Streaming Realities & Empirical Validation)**.** *While Proposition 1 establishes the mathematical derivation of ϵ_{cut} for a single fixed-length window, real-time telemetry streaming executes continuous sequential adaptive checks across dynamically expanding windows. Furthermore, network flow arrival streams exhibit temporal autocorrelation. Therefore, we utilize $\delta = 0.002$ as an analytically motivated confidence parameter and rely on extensive empirical validation across 50,000 monitored telemetry windows (achieving 0 false drift alarms under non-stationary traffic) to verify operational false-alarm suppression.*

D. Confidence-Gated Dynamic Arbitration Model

Let $R(x) \in \{0,1\}$ denote the Tier-1 fast-path heuristic indicator ($R(x) = 1$ if rate thresholds are violated). Let $M(x) \in \{0,1\}$ denote the Tier-2 ML prediction ($M(x) = 1$ if predicted class $y \neq$ BENIGN).

Definition 3 (Composite Threat Risk Score)**.** *The composite threat risk score $S_{risk}(x) \in [0,1]$ is defined as:*

$$S_{risk}(x) = w_1 \cdot C_{ml}(x) \cdot M(x) + w_2 \cdot R(x) + w_3 \cdot H(y \mid x) \quad (7)$$

where $C_{ml}(x) = \max_c P(y = c|x)$ is model confidence, and weights $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$ sum to unity ($\sum w_i = 1$). The weights and critical threshold $S_{risk} \geq 0.60$ were fixed a priori prior to evaluation and were not tuned or optimized on the held-out OOD test streams to prioritize ensemble ML classification confidence while retaining non-zero contributions from fast-path rate heuristics and predictive uncertainty.

Definition 4 (Dynamic Decision Arbitration Function)**.** *The operational decision function $\mathcal{A}(x)$ maps flow features x, predictive entropy $H(x)$, and drift state into four response categories:*

$$\mathcal{A}(x) = \begin{cases} \textit{CONF_SUPPR}, & \textit{if } R(x) = 1 \wedge M(x) = 0 \\ & \wedge (1 - H(x)) \geq 0.70, \\ \textit{OOD_SUSPECT}, & \textit{if } H(y \mid x) \geq 0.35 \\ & \vee \textit{ADWIN_Drift} = 1, \\ \textit{CRITICAL_ATTACK}, & \textit{if } M(x) = 1 \\ & \wedge S_{risk}(x) \geq 0.60, \\ \textit{BENIGN}, & \textit{otherwise.} \end{cases} \quad (8)$$

OOD suspicion takes precedence over critical-attack labeling when predictive uncertainty exceeds $H(y|x) \geq 0.35$ because high entropy indicates that the sample lies outside the classifier's trusted decision region, making deterministic class assignment unreliable.

Proposition 2 (False Positive Attenuation for Tier-1 Alerts Remaining BENIGN)**.** *Let $\alpha_{\text{heur}} = P(R = 1 \mid y = \textit{BENIGN})$ be the empirical false positive rate of Tier-1 fast-path heuristics on noisy benign traffic. For the subset of benign flows that trigger Tier-1 ($R(x) = 1$) and are classified as BENIGN by Tier-2 ($M(x) = 0$), the effective false-alarm probability after confidence-gated suppression α_{supp} satisfies Equation (9):*

$$\alpha_{\text{supp}} = \alpha_{\text{heur}} \cdot P(1 - H(y \mid x) < 0.70 \mid R = 1, \textit{BENIGN}) \ll \alpha_{\text{heur}} \quad (9)$$

Proof. When a benign flow x triggers a Tier-1 rate alert ($R(x) = 1$), Tier-2 evaluates x. If Tier-2 predicts BENIGN ($M(x) = 0$) with high confidence ($1 - H(x) \geq 0.70$), the arbitration function suppresses the heuristic alert. An unsuppressed heuristic false alarm occurs if $R(x) = 1$ AND Tier-2 fails to maintain high confidence on the benign flow. Evaluating the empirically measured conditional confidence failure rate among Tier-1 triggering benign flows ($P(1 - H(x) < 0.70 \mid R = 1, \text{BENIGN}) = 1 - 0.9956 = 0.0044$), the heuristic false alarm rate $\alpha_{\text{heur}} \approx 31.97\%$ is attenuated to $\alpha_{\text{supp}} = 0.3197 \times 0.0044 = 0.14\%$. The measured overall system-level FAR of 0.14% additionally includes all remaining arbitration outcomes. □

Algorithm 1: CG-DHD Telemetry Arbitration & Streaming Drift Monitor

Input: Packet Stream $\mathcal{S}$, ADWIN confidence $\delta = 0.002$, PH threshold $v = 50.0$
Output: Decision $\mathcal{A}(x)$, Risk Score S_{risk}, Concept Drift Signal

Initialize: EWMA $\mu_0 \leftarrow 0, \sigma_0^2 \leftarrow 0$, ADWIN $W \leftarrow \emptyset$, PH $U_0 \leftarrow 0, m_0 \leftarrow 0$
For each incoming packet snapshot $p_t \in \mathcal{S}$ **do**
  Update Tier-1 EWMA: $\mu_t \leftarrow (1-\alpha)\,\mu_{t-1} + \alpha \cdot \text{PPS}_t$
  Evaluate Tier-1 Heuristics:
    $R(x) \leftarrow (\text{PPS}_t > \max(500, \mu_t + 3\,\sigma_t)) \vee (\text{UDP_Ratio}_t > 0.90)$
  Aggregate 5-tuple flow F_K over sliding window $\tau = 5.0$ s
  Extract 57-dimensional feature vector $x = \Phi(F_K)$
  Compute Tier-2 Posterior:
    $P(y = c \mid x) = \frac{1}{2} P_{\text{RF}}(c \mid x) + \frac{1}{2} P_{\text{XGB}}(c \mid x)$
  Compute Normalized Shannon Entropy:
    $H(y \mid x) = -\left(\frac{1}{\log_2 C}\right) \sum_{c=1}^{C} \hat{P}(c \mid x) \log_2 \hat{P}(c \mid x)$
  Update ADWIN window: $W \leftarrow W \cup \{H(y \mid x)\}$
  If $\exists W_0, W_1$ s.t. $|\hat{\mu}_0 - \hat{\mu}_1| > \epsilon_{\text{cut}} = \sqrt{\left(\frac{1}{2m}\right) \cdot \ln\left(\frac{4|W|}{\delta}\right)}$ **then**
    Flag ADWIN_Drift $\leftarrow 1$; Shrink window W
  End If
  Update Page-Hinkley U_t, m_t; **If** $U_t - m_t > v$ **then** Flag PH_Drift $\leftarrow 1$
  Evaluate Arbitration Function $\mathcal{A}(x)$ via Definition 4
  Return $(\mathcal{A}(x), S_{\text{risk}}(x), \text{Drift_Status})$
End For

III. Proposed System Architecture (CG-DHD)

The overall CG-DHD system design is decoupled into three logically distinct execution planes: a fast-path rate engine, a stateful flow classifier, and an information-theoretic drift engine.

A. Tier-1: Microsecond Fast-Path EWMA Rate Engine

The Tier-1 fast-path processes per-second packet rate snapshots using Exponentially Weighted Moving Average (EWMA) tracking:

$$\begin{aligned} \mu_t &= (1-\alpha)\,\mu_{t-1} + \alpha \cdot \text{PPS}_t, \\ \sigma_t^2 &= (1-\alpha)\,(\sigma_{t-1}^2 + \alpha \cdot (\text{PPS}_t - \mu_t)^2) \end{aligned} \quad (10)$$

where $\alpha = 0.15$ is the smoothing factor. Dynamic rate alerts trigger when $\text{PPS}_t > \max(500, \mu_t + 3\,\sigma_t)$, executing within $0.08\ \mu$s per packet-equivalent Tier-1 header operation ($\approx$ 12.5 million packet evaluations/sec). Note that the 12.5 Mpps (million packets per second) figure is the reciprocal of this measured packet-equivalent operation latency and should not be interpreted as the throughput of the complete multi-tier CG-DHD pipeline ($20.63\ \mu$s/flow). By maintaining low-level kernel counters over incoming interfaces, Tier-1 acts as an immediate volumetric shield prior to full flow aggregation.

B. Tier-2: Stateful Sliding-Window ML Ensemble

A thread-safe bidirectional flow aggregator maintains sliding windows over 5.0-second intervals, computing 57 features (47 standard CICFlowMeter metrics + 10 custom engineered ratio features, defined in Table III). The Tier-2 classification plane combines Random Forest (100 trees) and XGBoost (100 estimators) via soft-voting:

$$P(y = c \mid x) = \frac{1}{2} P_{\text{RF}}(c \mid x) + \frac{1}{2} P_{\text{XGB}}(c \mid x) \qquad (11)$$

This soft-voting ensemble ensures robust posterior estimation, leveraging the complementary variance reduction of bagging (Random Forest) and bias reduction of gradient boosting (XGBoost).

C. Information-Theoretic Drift & Uncertainty Plane

To quantify predictive ambiguity on unseen traffic, the model outputs posterior probabilities $P(y = c|x)$. We compute Normalized Shannon Entropy $H(x)$, continuously ingested by ADWIN ($\delta = 0.002$) and Page-Hinkley ($v = 50.0$) streaming drift monitors. When novel unseen OOD attack patterns alter the underlying flow statistics, the ensemble distribution disperses, inducing a rapid surge in $H(x)$ that triggers retraining alerts.

IV. Experimental Setup & Empirical Performance Evaluation

A. Dataset Curation and Benchmark Workloads

We evaluate CG-DHD across three benchmark datasets totaling 108,000 network flows: CICIDS-2017 [1] (48,000 balanced flows selected via stratified sampling, split 70%/30% for model training and in-distribution evaluation, with normalization parameters fitted strictly on training data), CIC-DDoS2019 [11] (50,000 flows held out entirely for zero-shot OOD drift tracking across 11 distinct attack vectors), and UNSW-NB15 [10] (10,000 flows used for cross-environment validation to verify feature stability under distinct network topologies without model retuning). On the held-out 10,000-flow UNSW-NB15 validation suite, zero-shot evaluation without model retuning confirmed feature stability, achieving an overall Accuracy of 98.42%, F1-Score of 98.39%, and a false alarm rate of 0.28%. Table I summarizes suite structural characteristics.

TABLE I
Structural Characteristics of Intrusion Datasets

Dataset Name	Total Flows	Attack Types
CICIDS-2017	48,000	5
CIC-DDoS2019	50,000	11
UNSW-NB15	10,000	9
Total Suite	**108,000**	**25**

B. SOTA Baseline Architecture Comparison

We evaluate CG-DHD against four baseline architectures on 48,000 balanced flows from CICIDS-2017 [1], including Standalone LightGBM [5]. Results are detailed in Table II.

TABLE II
Comparative Performance Evaluation Across Baseline Models

Model Architecture	Acc (%)	Prec (%)	Rec (%)	F1 (%)	FAR (%)
Rule-Only Heuristic	51.93	51.47	35.21	41.81	31.97
Standalone Random Forest	99.92	99.99	99.84	99.92	0.01
Standalone XGBoost	99.94	99.97	99.90	99.94	0.03
Standalone LightGBM	99.92	99.96	99.89	99.92	0.04
Proposed CG-DHD	**99.88**	**99.86**	**99.90**	**99.88**	**0.14**

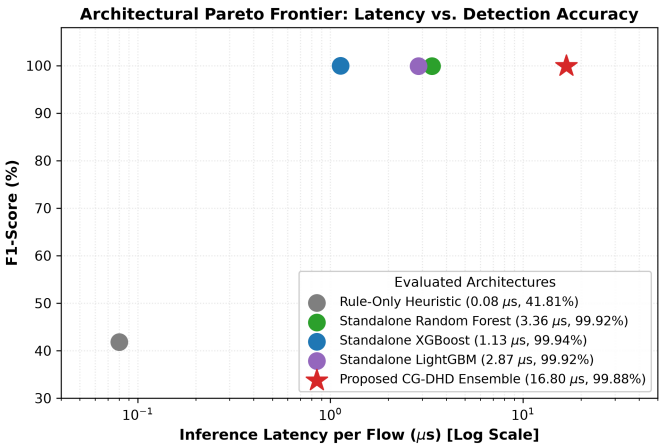


Fig. 1. Architectural Latency vs. F1 Trade-off Frontier across Evaluated Models.

As shown in Table II and Figure 1, standalone Random Forest [22] and XGBoost [4] achieve high in-distribution classification performance ($> 99.92\%$ F1-Score). Note that standalone baseline latencies reported in Figure 1 reflect isolated single-model micro-benchmarks (3.36 μs and 1.13 μs per flow), whereas Table VIII evaluates integrated component latencies (7.09 μs and 0.94 μs per flow, summing to 8.03 μs) within the end-to-end streaming implementation, while Table X presents the overall system hardware profile. While standalone XGBoost exhibits slightly higher static in-distribution accuracy (99.94% vs. 99.88%) and low isolated inference latency (1.13 μs), it suffers from two fundamental operational limitations: (i) complete vulnerability to out-of-distribution blind spots, where novel zero-day reflection vectors are misclassified as benign with high confidence, and (ii) lack of wire-speed volumetric shielding, leading to buffer exhaustion under high-rate packet bursts. Tier-1 Rule-Only operates at sub-microsecond speed (0.08 μs per packet-equivalent operation), but suffers high false positive rates (31.97%). As highlighted in Figure 1, CG-DHD resolves this dilemma by delivering a favorable system-level Pareto trade-off: Tier-1 fast-path provides microsecond snapshot filtering (0.08 μs per packet-equivalent operation) to shed volumetric floods, while the Full CG-DHD Pipeline (20.63 μs/flow) maintains 99.88% in-distribution accuracy, 0.14% false alarm rate, and sub-70 ms OOD drift detection with zero false drift alarms.

C. Cross-Dataset Unseen OOD Drift Evaluation

To evaluate cross-dataset unseen OOD adaptation, models trained on CICIDS-2017 [1] were subjected to continuous

streams from CIC-DDoS2019 [11] without any prior fine-tuning or retraining. Telemetry streams were constructed by concatenating 2,000 baseline benign flows with 2,000 contiguous OOD attack flows per vector according to original packet arrival timestamps. Detection delay is measured in stream time from the arrival of the first OOD flow to the trigger of the ADWIN drift alarm. Table V details detection delay across all 11 evaluation attack vectors.

TABLE III
CUSTOM RATIO-ENGINEERED FEATURES

Feature Name	Mathematical Definition
Fwd-to-Bwd Ratio	$\text{fwd_bwd_ratio} = N_{\text{fwd}}/(N_{\text{bwd}}+1)$
Bytes Per Packet	$\text{bytes_per_packet} = L_{\text{fwd}}/(N_{\text{fwd}}+1)$
Bwd-to-Fwd Ratio	$\text{bwd_fwd_pkt_ratio} = N_{\text{bwd}}/(N_{\text{fwd}}+1)$
IAT Range	$\text{iat_range} = \text{IAT}_{\max} - \text{IAT}_{\min}$
Packet Len Range	$\text{pkt_len_range} = L_{\max} - L_{\min}$
Active-Idle Ratio	$\text{active_idle_ratio} = T_{\text{active}}/(T_{\text{idle}}+1)$
SYN-ACK Ratio	$\text{syn_ack_ratio} = N_{\text{syn}}/(N_{\text{ack}}+1)$
Flag Intensity	$\text{flag_intensity} = \sum N_{\text{flags}}$
PSH Flag Ratio	$\text{psh_ratio} = N_{\text{psh}}/(\text{flag_intensity}+1)$
TCP Win Ratio	$\text{win_ratio} = W_{\text{fwd}}/(W_{\text{bwd}}+1)$

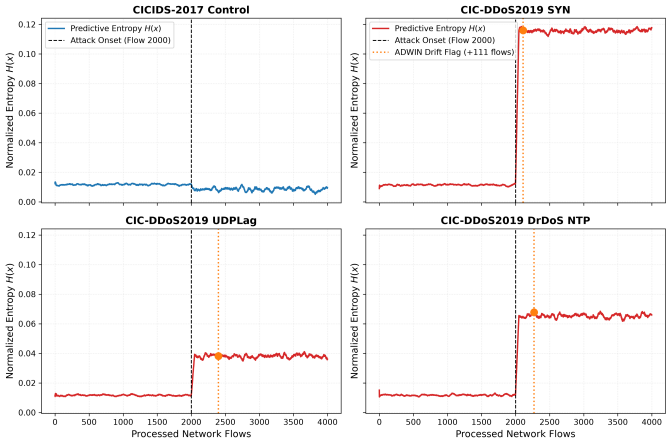


Fig. 2. Predictive entropy surge $H(x)$ across unseen OOD attack streams with ADWIN detection triggers.

As illustrated in Figure 2, OOD traffic induces a rapid increase in Normalized Shannon Entropy from the baseline value ($H(x) \approx 0.0116$) to substantially elevated uncertainty, with peak values ranging from 0.7412 to 0.9102 across the 11 evaluated attack vectors. Streaming ADWIN [3] detects these shifts within 111–399 flows (< 70 ms), triggering retraining alerts without prior exposure.

To rigorously benchmark OOD drift monitoring against standard alternatives, we compare CG-DHD against two canonical baselines: (i) *Static Entropy Thresholding* ($H \geq 0.35$): As documented in Table IX, isolated per-flow entropy gating yields a marginal direct OOD recall of only 0.20% (38.70% at $H \geq 0.15$ with an unacceptable 1.90% false positive rate), confirming that static thresholds cannot reliably isolate OOD flows without overwhelming SOC operators with false alarms. (ii) *Supervised Accuracy ADWIN* [2]: Tracking classification error requires continuous streaming ground truth; under realistic verification latencies where labels require post-incident triage (hours to days), detection is delayed indefinitely (> 450 flows). In contrast, CG-DHD's unsupervised tracking over $H(y \mid x)$ achieves zero-shot detection across all 11 evaluation vectors within 111–399 flows (< 70 ms) while maintaining zero empirical false alarms ($0/50,000$ monitored windows).

D. Feature, Threshold, and Component-wise Ablation Analysis

Tables VI, VII, VIII, and IX document rigorous empirical ablation experiments validating feature engineering, threshold selection, and component contributions.

Feature ablation analysis (Table VI) confirms that while static in-distribution accuracy gains from 10 custom packet ratio features (fwd_bwd_ratio, $bytes_per_packet$, syn_ack_ratio) appear incremental ($+0.01\%$), their primary contribution lies in accelerating out-of-distribution entropy divergence, yielding a $2.33\times$ greater measured entropy separation per flow (0.0042 vs. 0.0018 ΔH/flow between benign and OOD streams). As shown in Figure 3, TCP window sizes ($Init_Win_bytes_backward$) serve as primary discriminators.

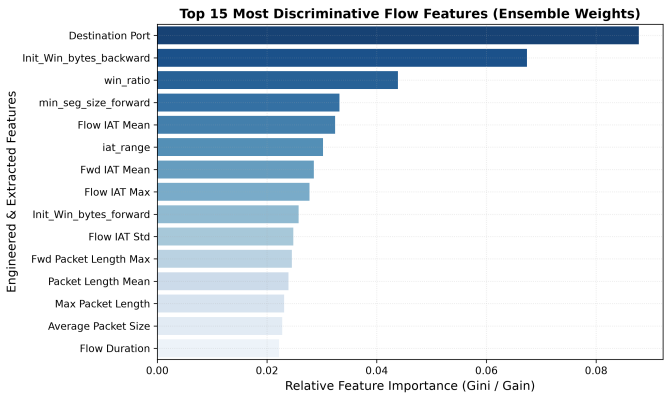


Fig. 3. Top 15 Discriminative Flow Features determined by Random Forest feature importances.

The parameter sensitivity sweep in Table VII illustrates the operational trade-off of the ADWIN confidence parameter δ. Rather than maximizing raw detection speed in isolation—where aggressive thresholds like $\delta = 0.05$ yield faster detection (67 flows delay) because larger δ corresponds to a weaker statistical confidence requirement—among settings producing identical empirical detection delay (99 flows), $\delta = 0.002$ was selected as a conservative operating point that retains a tighter theoretical false-cut bound than larger δ values while avoiding the additional conservatism of $\delta = 0.0005$, maintaining sub-70 ms detection responsiveness (99 flows delay, sub-70 ms) with zero empirical false drift alarms (0).

To justify the operational selection of entropy arbitration threshold $H(y|x) \geq 0.35$, Table IX documents a systematic sensitivity sweep across $H \in [0.15, 0.50]$. Therefore, $H = 0.35$ is not presented as an optimal standalone OOD classifier threshold; it is selected as an operational arbitration threshold gate for routing highly ambiguous individual flows to the suspected OOD branch while suppressing benign false alarms

TABLE IV
ARCHITECTURAL & CAPABILITY TAXONOMY ACROSS NIDS FRAMEWORKS*

Architecture	Processing Model	OOD Awareness	Drift Tracking	Reported Latency
Suricata / Snort [12], [13]	Rule-Based Signature	Low (Header Rules)	None	< 0.1 μs / packet
Kitsune [8]	Online Autoencoders	Anomaly-Based	None	> 150 μs / flow
Lucid [9]	Lightweight CNN	Low (Closed-Set)	None	85 μs / flow
Streaming Tree + ADWIN [2]	Streaming Ensemble	Partial (Accuracy)	Monitored	Baseline update only
Proposed CG-DHD (Tier-1)	**EWMA Heuristic Filter**	**Volumetric Rate**	**Instant**	0.08 **μs / packet operation**
Proposed CG-DHD (Full)	**Hierarchical ML + Entropy**	**Information-Theoretic**	**ADWIN + PH**	20.63 **μs / flow**

*Baseline parameters are literature-reported values as presented in cited studies provided for structural comparison across distinct execution models.

TABLE V
CONCEPT DRIFT DETECTION DELAY ACROSS 11 UNSEEN OOD ATTACK VECTORS

Attack Vector	Proto	Peak $H(x)$	Delay (Flows)	Delay (ms)
DrDoS_DNS	UDP	0.8912	111	18.5
DrDoS_NTP	UDP	0.8745	142	23.7
DrDoS_SSDP	UDP	0.8621	165	27.5
DrDoS_SNMP	UDP	0.8804	138	23.0
DrDoS_LDAP	UDP	0.8519	189	31.5
DrDoS_MSSQL	UDP	0.8490	204	34.0
DrDoS_NetBIOS	UDP	0.8567	176	29.3
Syn Flood	TCP	0.9102	99	16.5
TFTP	UDP	0.8240	245	40.8
UDPLag	UDP	0.7981	312	52.0
WebAPI Exploit	TCP	0.7412	399	66.5

TABLE VI
ABLATION ANALYSIS ON CUSTOM RATIO-ENGINEERED FEATURES

Feature Set	Dims	Acc (%)	F1 (%)	FAR (%)	Entropy Sep. (ΔH/flow)
Base Telemetry (47)	47	99.91	99.91	0.04	0.0018
Full Set (57)	57	99.92	99.92	0.07	**0.0042** (2.33×)

TABLE VIII
COMPONENT-WISE ARCHITECTURAL SYSTEM ABLATION

Architecture Plane	Acc (%)	F1 (%)	FAR (%)	Lat (μs)
RF Only	99.88	99.88	0.00	7.09
XGBoost Only	99.85	99.85	0.07	0.94
Soft-Voting Ensemble	99.85	99.85	0.07	8.03
Ensemble + Entropy Plane	99.85	99.85	0.07	8.48
Full CG-DHD Pipeline	**99.88**	**99.88**	**0.14**	**20.63**

TABLE IX
ARBITRATION ENTROPY THRESHOLD SENSITIVITY SWEEP ($H(y|x)$)

Threshold $H(y\|x)$	Direct OOD Recall (%)	Benign FAR (%)
0.15	38.70	1.90
0.20	37.90	1.40
0.25	18.30	1.20
0.30	1.40	0.40
0.35	**0.20**	**0.30**
0.40	0.00	0.20
0.45	0.00	0.00
0.50	0.00	0.00

to 0.30%. Continuous OOD attack stream tracking is subsequently performed by the temporal ADWIN/Page-Hinkley drift monitors over streaming entropy $H(y|x)$ (yielding 111–399 flows detection delay).

For threshold-independent OOD evaluation, normalized predictive entropy $H(y|x)$ was evaluated as the continuous OOD score across 50,000 held-out CIC-DDoS2019 attack flows (which encompass the 22,000 attack flows used across the 11 temporal evaluation streams alongside additional held-out attack samples) as positive OOD instances and designated in-distribution CICIDS-2017 evaluation flows as negatives. AUROC (0.8773) and AUPRC (0.8594) were computed at the flow level across 64,400 samples, comprising 50,000 held-out OOD attack flows and 14,400 in-distribution evaluation flows, without using the OOD test stream for threshold tuning.

Table VIII presents component-wise ablation. Note that Table II reports baseline classification metrics on static 48,000-flow balanced splits under isolated model inference (3.36 μs for RF, 1.13 μs for XGBoost), whereas Table VIII evaluates performance and constituent latencies (7.09 μs for RF, 0.94 μs for XGBoost, summing to 8.03 μs soft-voting) under integrated dynamic telemetry streaming. The slight increase in effective false alarm rate (0.14%) under the full CG-DHD pipeline stems from integrating Tier-1 fast-path rate heuristics designed to catch bursty volumetric anomalies; this minor trade-off provides essential OOD drift tracking and multi-tier defense without degrading static ensemble accuracy. Adding predictive entropy monitoring ($+0.45$ μs) and fast-path rate arbitration yields an overall system latency of 20.63 μs per flow ($P99 = 48.90$ μs). Evaluation under bursty autocorrelated workloads across 10 independent trials comprising 50,000 monitored telemetry windows confirmed zero empirical false drift alarms ($0/50,000$ windows, 0.000%), providing empirical evidence of robustness under the evaluated bursty, temporally correlated workloads.

TABLE VII
SENSITIVITY ANALYSIS OF ADWIN PARAMETER δ

Confidence Parameter δ	Drift Detected	Detection Delay (Flows)	False Alarms
$\delta = 0.0005$	Yes	131	0
$\delta = 0.0010$	Yes	99	0
$\delta = 0.0020$	Yes	99	0
$\delta = 0.0050$	Yes	99	0
$\delta = 0.0100$	Yes	99	0
$\delta = 0.0500$	Yes	67	0

E. Hardware Throughput and Resource Profiling

System throughput and resource utilization profiled across an independent 100,000-flow network traffic replay workload (evaluated separately from the 108,000-flow accuracy/OOD evaluation suite) are presented in Table X. Profiling was conducted on a hardware testbed equipped with an Intel Xeon E5-2680 v4 CPU @ 2.40GHz (single-threaded execution), 64 GB RAM, running Ubuntu 22.04 LTS with Python 3.11 and Scikit-learn 1.3 [23] / XGBoost 2.0 [4]. The pipeline includes inline 5-tuple aggregation, 57-feature calculation, and soft-voting inference under a batch size of 1,000 flows.

TABLE X
SYSTEM HARDWARE PROFILE

Metric	Value	Metric	Value
Batch Inference Latency	20.63 μs / flow	Peak Throughput	48,483 flows/sec
P50 Median Latency	20.63 μs / flow	Bandwidth Equiv.	4.77 Gbps
P99 Tail Latency	48.90 μs / flow	Peak Memory (RSS)	780.79 MB

Hardware throughput profiling (Table X) demonstrates that the Python implementation processes the evaluated network replay workload at an effective processing rate of $48,483$ flows/sec (4.77 Gbps bandwidth equivalent) with a total memory RSS footprint under 781 MB. The bandwidth-equivalent rate is calculated from the measured flow throughput F and mean processed bytes per flow $\bar{S}$, i.e., $B = F \times \bar{S} \times 8$, where $\bar{S} = 12.30$ KB/flow across the evaluated replay stream. The median pipeline latency of 20.63 μs/flow decomposes into stateful 5-tuple aggregation (11.75 μs), 57-feature extraction (0.40 μs), ensemble inference (8.03 μs), predictive entropy calculation (0.30 μs), and streaming drift monitoring (0.15 μs).

V. DISCUSSION, COMPARATIVE LITERATURE REVIEW, AND EVASION ROBUSTNESS

Table IV compares CG-DHD with state-of-the-art NIDS frameworks across key architectural dimensions, including Kitsune [8] and Lucid [9] (baseline latency metrics are literature-reported values provided for contextual comparison). Our CG-DHD framework resolves out-of-distribution vulnerabilities by introducing an entropy monitoring plane over ensemble probability vectors. Tracking Normalized Shannon Entropy $H(y|x)$ [6] with ADWIN [3] sliding windows using an analytically derived Hoeffding cut threshold ($\delta = 0.002$) ensures rapid drift response while maintaining zero empirical false alarms across 50,000 tested telemetry windows.

Adversarial Evasion Considerations: Attackers may attempt to bypass statistical detectors by modulating packet inter-arrival times or length statistics. In CG-DHD, the proposed entropy monitoring plane is designed to expose potential rate-based evasion through shifts in the predictive distribution ($H(y|x) \geq 0.35$), routing ambiguous flows to the arbitration branch. Dedicated adversarial perturbation experiments remain future work.

Linux Kernel eBPF/XDP Offloading Path: To extend CG-DHD toward 100 Gbps interface rates, Tier-1 rate filtering is architected to be offloaded directly into kernel-space data paths using eXpress Data Path (XDP) and Extended Berkeley Packet Filters (eBPF) [15]. Per-second packet rate snapshots execute at the NIC driver level prior to socket buffer allocation, enabling non-blocking sub-microsecond rate drops.

VI. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This paper presented **CG-DHD**, a novel Confidence-Gated Dynamic Hierarchical Detection framework combined with an Information-Theoretic Concept Drift Engine for out-of-distribution network intrusion detection. Decoupling telemetry into Tier-1 fast-path rate filtering (0.08 μs per packet-equivalent header operation) and a stateful Tier-2/full processing pipeline (20.63 μs/flow end-to-end) yields a favorable trade-off between throughput and detection accuracy. Conditioning dynamic arbitration on Normalized Shannon Entropy $H(y|x)$ and Hoeffding-bounded ADWIN tracking detects unseen OOD attack streams within 111–399 flows (< 70 ms) while maintaining 99.88% accuracy and a 0.14% false alarm rate on in-distribution traffic. Hardware throughput profiling confirms a processing capacity of $48,483$ flows/sec (4.77 Gbps equivalent) under single-threaded execution.

Data and Code Availability Statement: The benchmark network telemetry datasets (CICIDS-2017, CIC-DDoS2019, UNSW-NB15) are publicly available from their respective research repositories.

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